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EE-102, Section 003

Lab 1 : Introduction to Digital Oscilloscopes

Experiment 1

27.09.2023

AIM:

The main purpose of our experiment is to investigate how digital oscilloscope and signal generator work and get familiar working with such equipments. Moreover, our secondary purpose is learing to design circuits by using a breadboard.

METHODOLOGY and RESULTS

In the experiment, the oscilloscope and signal generator were the tools we especially use. Our experiment contains 6 parts :

PART 1

Initially, we need to compensate the oscilloscope probe before the experiment because it can cause errors on our measurements. The oscilloscope probe was selected the attenuation factor of 10X throught this experiment. We then connected the oscilloscope probe to the prob comp and the ground part of the oscilloscope to compensate for this. Using the probe signal, which is the square waveform produced by the oscilloscope, we can set the inner circuity of our probe to observe square waveform on our oscilloscope . If it is not appropriately adjusted there are two possibilities happen : under-compensation **(Figure 1.1)** and over-compensation **(Figure 1.2)** which will cause change our measurements . If it is accurately compensated a waveform similar to **(Figure 1.3)** can be seen.

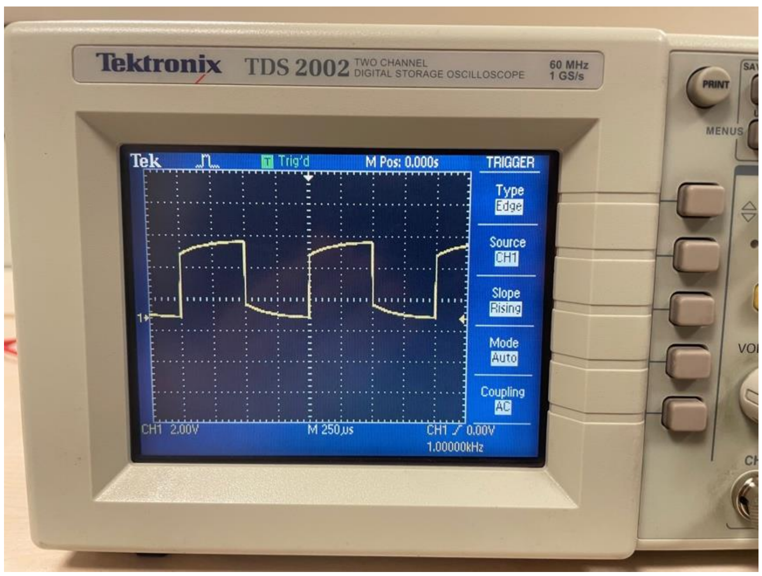

Figure 1.1 : Under compensation

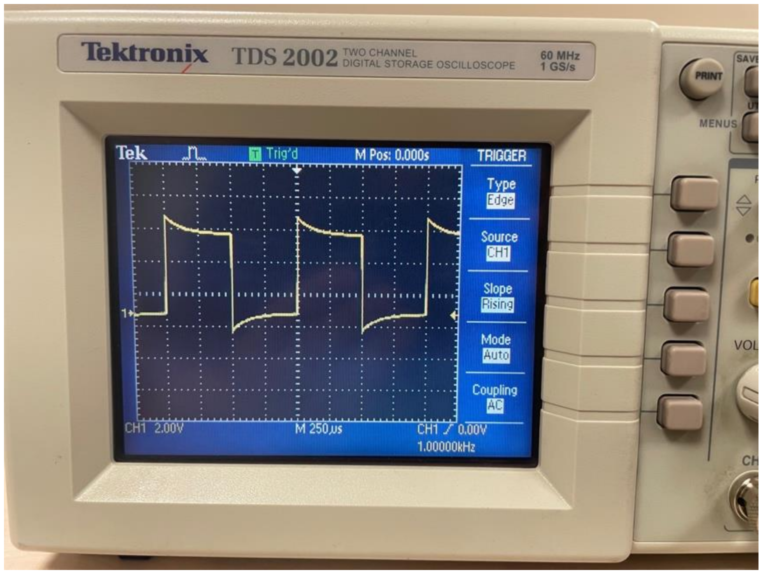

Figure 1.2 : Over compensation

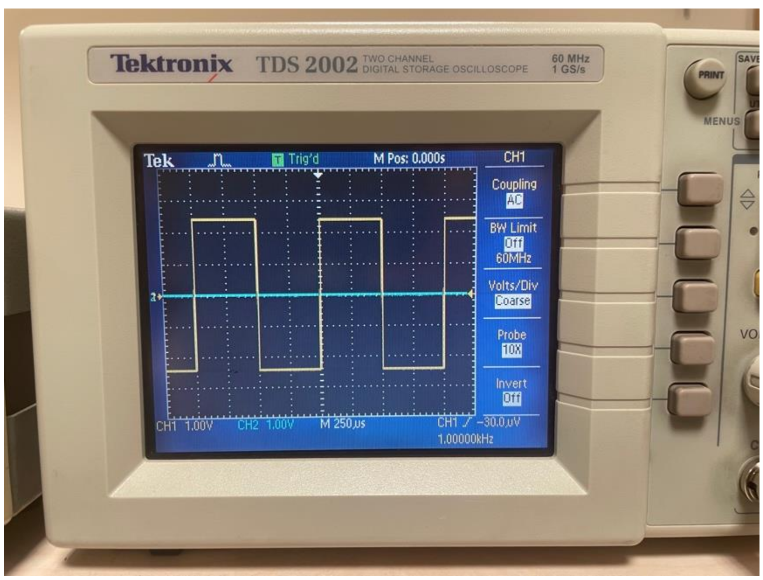

Figure 1.3 : Proper compensation

PART 2

We established a connection between the signal generator and our oscilloscope using a coaxial BNC cable in our first experiment. We then connected the function output of the signal generator to the Ch1 input on our oscilloscope. A 5V peak-to-peek sinusodial signal with a frequency of 1kHz was generated by the signal generator. Firstly, we used positive edge triggering as shown in **(Figure 2.1)** to see waveform as it is going from low voltage to high voltage. We then switched to negative edge triggering as in **(Figure 2.2)** to track the descending side of the waveform. It is important to note that both triggers display the same waveform but capture it at different times.

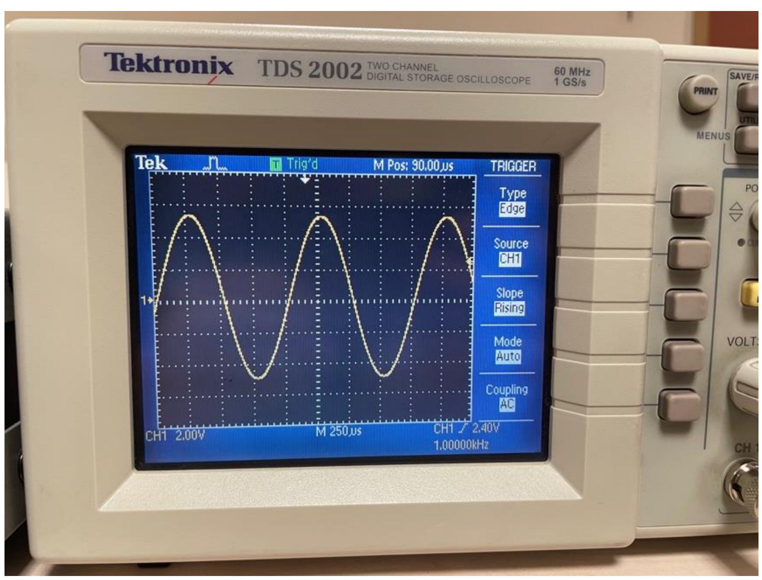

Figure 2.1 : Positive edge triggering

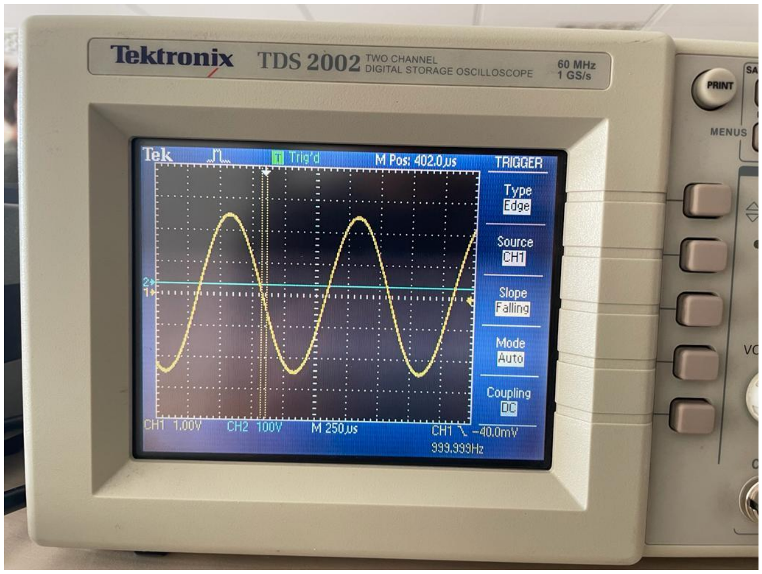

Figure 2.2 : Negative edge triggering

PART 3

We changed our signal to be a triangular wave with 2kHz frequency and 1V peak to amplitude the stable form is displayed as **(Figure 3.1)** . After observing that waveform on oscilloscope , we turn the trigger knob to observe its effects. It is observed that trigger knoob changes the trigger level , the point where the signal coincides with the origin of the oscilloscope is shifted according to the direction of rotation. The trigger level thus changes the time and place where trigger occurs. If the trigger points is outside the amplitude range, the signal seen on the oscilloscope as unstable **(Figure 3.2)**. The oscilloscope can be triggered with increasing or decreasing voltage signal. These triggers called positive and negative edge triggering as we see in the second part.

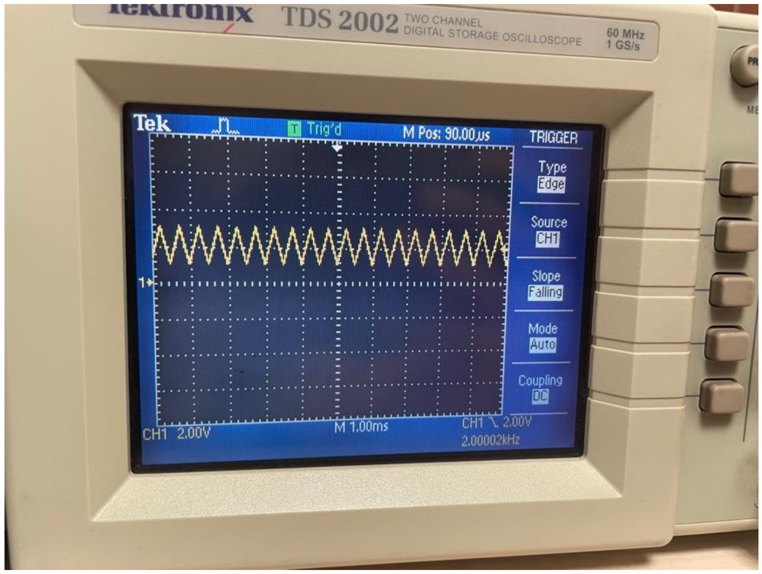

Figure 3.1 : Stable waveform with triggering

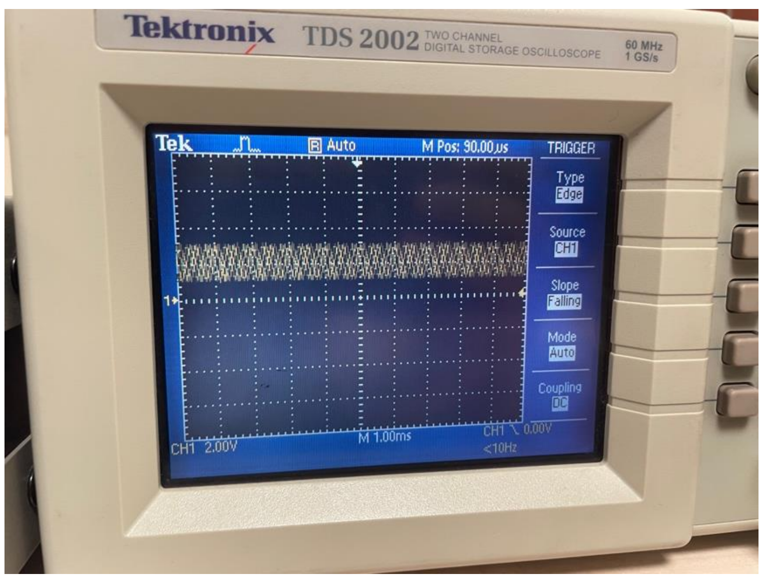

Figure 3.2 : Unstable waveform without triggering

PART 4

We converts digitals signals to analog signals by using the system digital to analog converter (DAC) . It takes discrete values and produces a continious value output. DAC's are widely used in a variety of applications where digital systems required interface with analog components such as headphones. Analog to digital converter (ADC) is a system which converts analog signals into digital signals. There are some areas ADC can be used such as microphones and cameras which have analog inputs. Moreover, oscilloscope uses ADC to ample and digitize the incoming analog signal therefore, oscilloscope measure and display analog voltage waveforms as digital on its display. In this part of the experiment we adjust signal to be a square wave with 5 kHz frequency and 1V peak-to-peak amplitude. We observe the difference between various acqusition modes "sample","peak","detect",and "average" on the oscilloscope. We can observe basic mode on **(Figure 4.1)** which is "sample" mode. In sample mode each sample of the input signal acquired and displayed by oscilloscope and it detecets the waveform immediately without repetitive sampling. However, it may be observed that it displays rough and inexplicit edges and it is not the most useful when very precise measurements are taken into consideration. In **(Figure 4.2)** we use peak detect mode. There is no big difference between **(Figure 4.1)** and **(Figure 4.2)** because our wave is very stable and has no high frequency. Nevertheless, if the signal is unstable or the frequency of wave is much higher, we measure the waveform more accurate thanks to peak detect mode, so , it is beneficial when you want to capture and analyze glitches or spikes. We use average mode with 16 averages and we can observe at **(Figure 4.3)** smoother and stable waveform with low noise level on oscilloscope display. Average mode provides a high-resolution waveform thanks to the average value taken from the sample points, but cannot detect spikes and pulses.

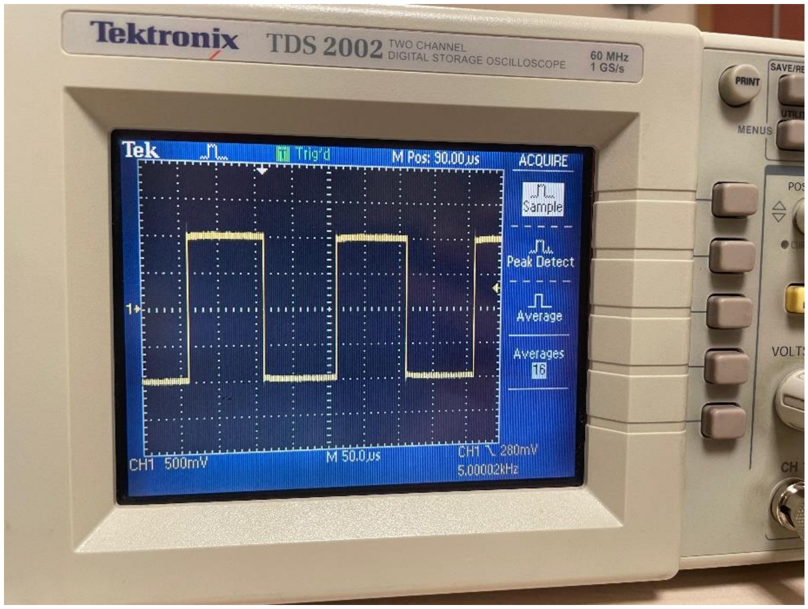

Figure 4.1 : Sample Mode

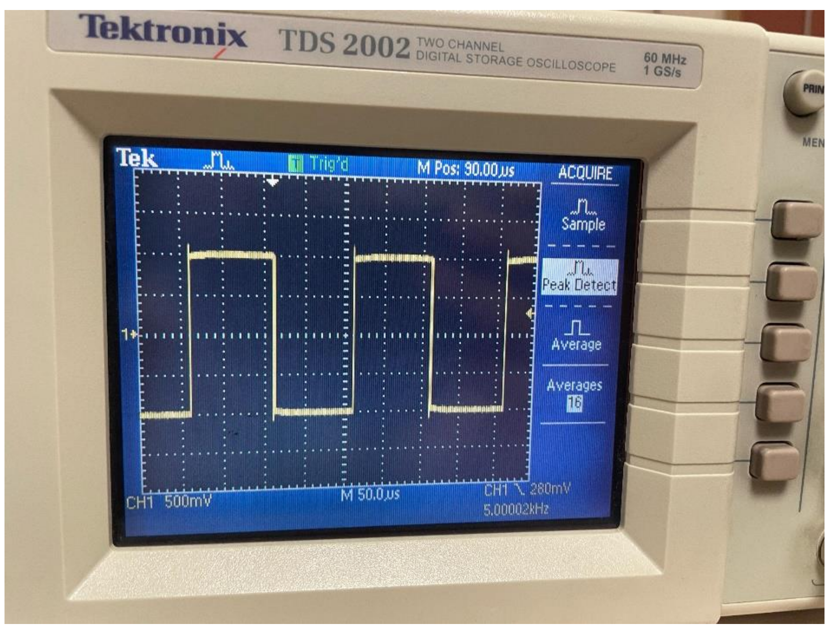

Figure 4.2 : Peak Detect Mode

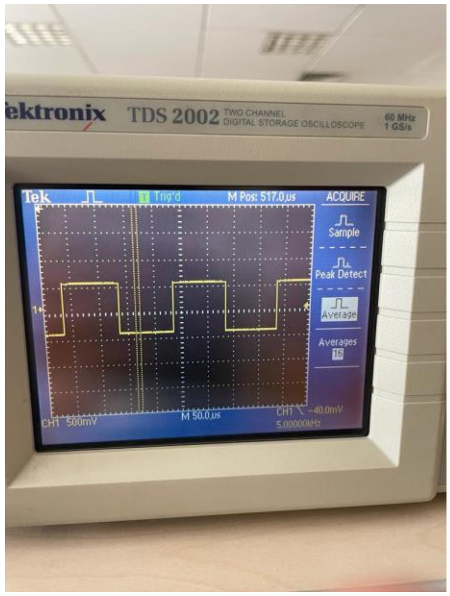

Figure 4.3 : Average Mode

PART 5

In this part of the experiment 2V peak-to-peak amplitude with a DC offset of 1V 1kHz sinusoidal signal was applied by the signal generator. When we use DC coupling to observe the waveform, we can see the signal is shifted 1V up from the horizontal axis **(Figure 5.1)**. When AC coupling is selected, we can only see the sinusodial oscillations of the waveform and the offset of the input signal is completely filtered and not present as in **(Figure 5.2)**.

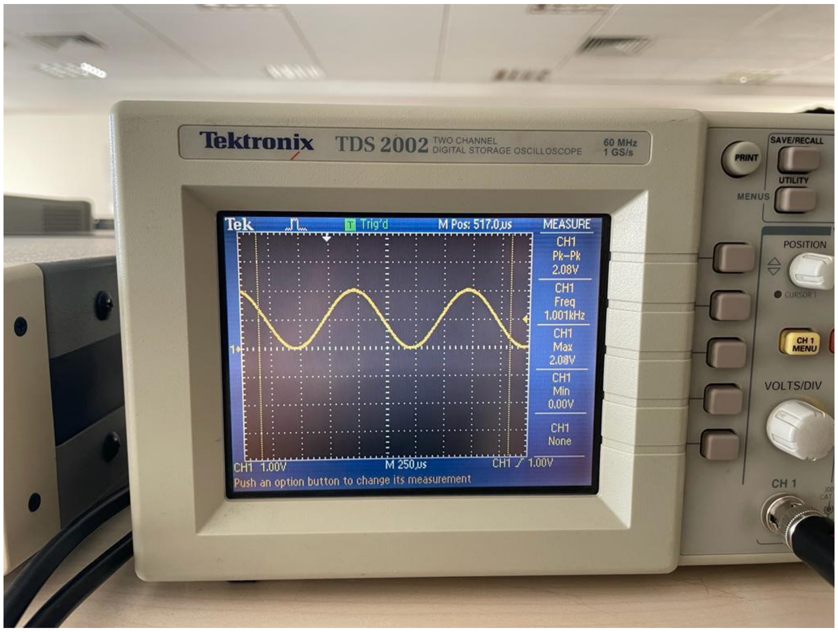

Figure 5.1 : DC coupling with offset

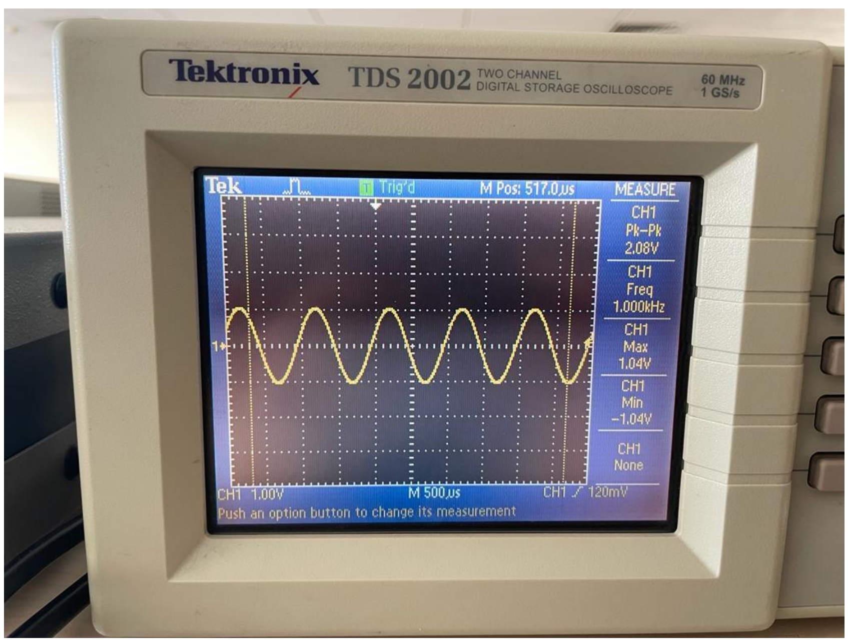

Figure 5.2 : AC coupling with offset

PART 6

In this part of the experiment we incorporate a breadboard into our experiment. Breadboard can be explained as as an array of connected circuit lines. We can design and construct circuits without soldering them. It consists of flat board with a grid of holes that are connected internally so we can insert or remove any circuit item easily. We start by making the circuit design in **(Figure 6.1)** on our breadboard. This AC circuit design contains a 1 μF capacitor and a 1 kΩ resistor. The completed circuit is looks as in **(Figure 6.2)** Then we connected 2 oscilloscope probes X and Y as described in **(Figure 6.1)** and this can be observed with the oscilloscope connections as shown in **(Figure 6.3).** Moreover, we set our signal as a sinusoidal signal with 1kHz frequency and 2V peak-to-peak voltage with 0 DC offset which seen on our oscilloscope as **(Figure 6.4)**. Next we will closer look at the phase difference of the two waves. We can observe the delay measured by oscilloscope as 26 microsecond between two waves. There is no need to calculate the phase difference manually because this model of oscilloscope could measure the phase difference which is -9.35 ° which is 0.163 rad in the experiment with 1kHz.

Now we change the frequency to 100kHz **(Figure 6.5)**. We can see the the delay between the signal X and Y is 17 nanoseconds and the phase difference is -0.61 ° which is -0.011 rad in the experiment with 100kHz. The phase difference between X and Y waveforms is much less when we use 100kHz compared to 1kHz. Therefore we can say using higher frequencies reduces the phase difference.

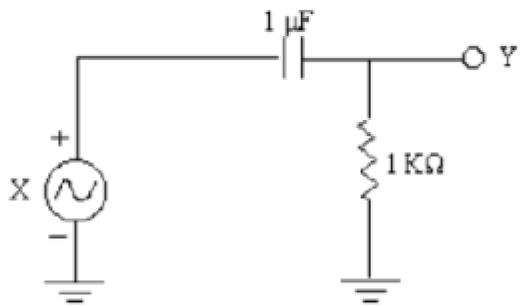

Figure 6.1 : Circuit Design

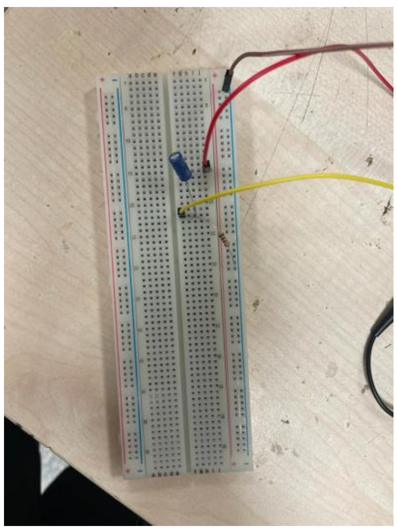

Figure 6.2 : Completed circuit on the breadboard

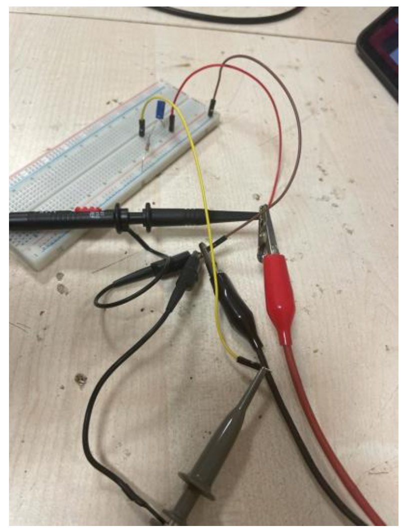

Figure 6.3 : Breadboard with oscilloscope connections

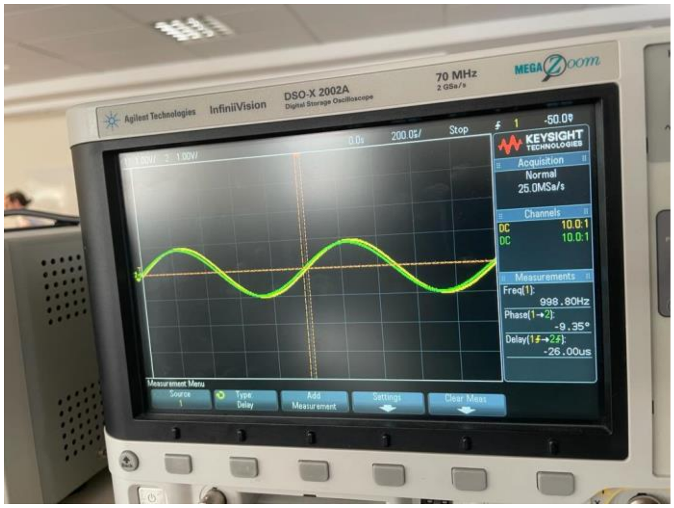

Figure 6.4 : 1kHz two signals with delay and phase difference values

Figure 6.5 : 100kHz signal with delay and phase difference values

CONCLUSİON

The purpose of this experiment was to learn how to use typical functions of function generator and oscilloscope. In general, no problems beyond our purpose or unexpected errors were encountered in the experiment. Predictable margins of error can be affected by factors such as the quality of the oscilloscope or oscilloscope probe. The circuit was set up on the breadboard as specified and the expected results was achieved. Moreover, while adjusting the amplitude of waves produced by the function generator, we had to enter ½ the peak to peak amplitude value in order to observe required value on the oscilloscope.